

Task 1: Develop a C program that demonstrates how a Linux operating system executes a command entered by a user that

1. Accept a Linux command as input.
2. Create a child process using `fork()`.
3. Execute the command in the child process using an appropriate `exec()` system call.
4. Allow the parent process to wait for the child using `wait()`.
5. Display the Process ID (PID) of both parent and child processes.

- a) Write a C code to create a process using `fork()` system call.
- b) Read the user input as shell command.
- c) Use a system call `exec()` to execute the shell command.
- d) Use a `wait()` system call for waiting a parent process.
- e) Display the PIDs of parent and child processes.

```
GNU nano 7.2 p2t1.c *
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>

int main()
{
    pid_t pid;
    char command[100];

    // b) Read user input as shell command
    printf("Enter a Linux command: ");
    scanf("%s", command);

    // a) Create a child process using fork()
    pid = fork();

    if (pid < 0)
    {
        printf("Fork failed!\n");
        exit(1);
    }

    else if (pid == 0)
    {
        // Child Process
        printf("\nChild Process\n");

        // e) Display child and parent PIDs
        printf("Child PID : %d\n", getpid());
        printf("Parent PID : %d\n", getppid());

        printf("Executing command: %s\n", command);

        // c) Execute command using exec()
        execlp(command, command, (char *)NULL);

        // This executes only if exec() fails
        printf("Exec failed!\n");
        exit(1);
    }

    else
    {
        // Parent Process
        printf("\nParent Process\n");

        // e) Display parent and child PIDs
        printf("Parent PID : %d\n", getpid());
        printf("Child PID : %d\n", pid);

        // d) Parent waits for child
        wait(NULL);

        printf("\nChild process completed.\n");
    }

    return 0;
}
```

```

v_gopinath@DESKTOP-3QLP53Q:~$ nano p2t1.c
v_gopinath@DESKTOP-3QLP53Q:~$ gcc p2t1 -o
gcc: error: missing filename after '-o'
v_gopinath@DESKTOP-3QLP53Q:~$ gcc p2t1.c -o p2t1
v_gopinath@DESKTOP-3QLP53Q:~$ ./p2t1
Enter a Linux command: ls

Parent Process
Parent PID : 1222
Child PID  : 1223

Child Process
Child PID   : 1223
Parent PID  : 1222
Executing command: ls
fork.c fork2 fork2.c fork_exec fork_exec.c hello.c myShell myShell.c p2t1 p2t1.c

Child process completed.
v_gopinath@DESKTOP-3QLP53Q:~$ ls
fork.c fork2 fork2.c fork_exec fork_exec.c hello.c myShell myShell.c p2t1 p2t1.c
v_gopinath@DESKTOP-3QLP53Q:~$ |

```

Task-2: Using Linux terminal commands (uname, lscpu, ps, top), investigate the relationship between hardware resources and operating system services. Prepare a report explaining how the OS abstracts CPU, memory, storage, and I/O devices.

```

v_gopinath@DESKTOP-3QLP53Q:~$ gcc p2t1.c -o p2t1
v_gopinath@DESKTOP-3QLP53Q:~$ ./p2t1
Enter a Linux command: uname

Parent Process
Child Process
Parent PID : 1272
Child PID  : 1273
Child PID  : 1273
Parent PID : 1272
Executing command: uname
Linux

Child process completed.
v_gopinath@DESKTOP-3QLP53Q:~$ lscpu
Architecture:            x86_64
CPU op-mode(s):          32-bit, 64-bit
Address sizes:            48 bits physical, 48 bits virtual
Byte Order:               Little Endian
CPU(s):                   4
On-line CPU(s) list:      0-3
Vendor ID:                AuthenticAMD
Model name:               AMD Ryzen 3 3200U with Radeon Vega Mobile Gfx
CPU family:               23
Model:                    24
Thread(s) per core:       2
Core(s) per socket:       2
Socket(s):                 1
Stepping:                  1
BogoMIPS:                  5190.09
Flags:                     fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ht syscall nx mmxext fxsr_opt pdpe1gb r
dtscp lm constant_tsc rep_good nopl xtopology tsc_reliable nonstop_tsc cpuid extd_apicid tsc_known_freq pni pclmulqdq ssse3 fma cx16 sse4
_1 sse4_2 movbe popcnt aes xsave avx f16c rdrand hypervisor lahf_lm cmp_legacy svm cr8_legacy abm sse4a misalignsse 3dnowprefetch osvw to
poext perfctr_core ssbd ibpb vmcall fsgsbase bmi1 avx2 smep bmi2 rdseed adx smap clflushopt sha_ni xsaveopt xsavec xgetbv1 clzero xsavee
rptr virt_ssbd arat npt nrip_save tsc_scale vmcb_clean flushbyasid decodeassists pausefilter pfthreshold v_vmsave_vmload

Virtualization features:
Virtualization:           AMD-V
Hypervisor vendor:        Microsoft
Virtualization type:       full
Caches (sum of all):
L1d:                       64 KiB (2 instances)
L1i:                       128 KiB (2 instances)
L2:                         1 MiB (2 instances)
L3:                         4 MiB (1 instance)
NUMA:
NUMA node(s):              1
NUMA node0 CPU(s):         0-3

```

```
NUMA node(s): 1
NUMA node0 CPU(s): 0-3
Vulnerabilities:
Gather data sampling: Not affected
Ghostwrite: Not affected
Indirect target selection: Not affected
Itlb multihit: Not affected
L1tf: Not affected
Mds: Not affected
Meltdown: Not affected
Mmio stale data: Not affected
Old microcode: Not affected
Reg file data sampling: Not affected
Retbleed: Mitigation; untrained return thunk; SMT vulnerable
Spec rstack overflow: Vulnerable; Safe RET; no microcode
Spec store bypass: Mitigation; Speculative Store Bypass disabled via prctl
Spectre v1: Mitigation; usercopy/swapgs barriers and __user pointer sanitization
Spectre v2: Mitigation; Retpolines; IBPB conditional; STIBP disabled; RSB filling; PBRSE-IBRS Not affected; BHI Not affected
Srbds: Not affected
Tsa: Not affected
Tsx async abort: Not affected
Vmscape: Not affected
```

v.gopinath@DESKTOP-3QLP53Q:~\$ ps

```
PID TTY TIME CMD
462 pts/0 00:00:01 bash
1295 pts/0 00:00:00 ps
```

v.gopinath@DESKTOP-3QLP53Q:~\$ top

```
top - 13:34:28 up 1:23, 1 user, load average: 0.00, 0.00, 0.00
Tasks: 25 total, 1 running, 24 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni,100.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 2880.6 total, 2330.1 free, 430.1 used, 194.3 buff/cache
MiB Swap: 1024.0 total, 1024.0 free, 0.0 used, 2450.5 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
1	root	20	0	24332	15400	11556	S	0.0	0.5	0:23.09	systemd
2	root	20	0	3180	2204	2072	S	0.0	0.1	0:00.76	init-systemd(Ub
8	root	20	0	3180	2044	1940	S	0.0	0.1	0:00.01	init
63	root	19	-1	50396	16748	15444	S	0.0	0.6	0:00.76	systemd-journal
109	systemd+	20	0	22416	14436	11940	S	0.0	0.5	0:01.00	systemd-resolve
121	root	20	0	35244	12200	9228	S	0.0	0.4	0:11.56	systemd-udev
138	root	20	0	8644	5492	4652	S	0.0	0.2	0:00.42	login
184	root	20	0	2888	1948	1820	S	0.0	0.1	0:00.53	chronyd-starter
194	root	20	0	4472	3032	2780	S	0.0	0.1	0:00.45	cron
196	message+	20	0	8816	5308	4612	S	0.0	0.2	0:00.82	dbus-daemon
210	root	20	0	44096	29752	14612	S	0.0	1.0	0:02.21	networkd-dispat
221	root	20	0	18448	9464	8172	S	0.0	0.3	0:00.81	systemd-logind
247	syslog	20	0	220548	5816	4516	S	0.0	0.2	0:02.18	rsyslogd
251	root	20	0	5492	2728	2540	S	0.0	0.1	0:00.24	agetty